

FCFS Simulation: Main Conclusions by Mechanism

May 2026 · Two-class discrete-event FCFS appointment simulation

This sheet summarizes the model-level conclusions from the FCFS simulation work.
The goal is to separate metric interpretation, behavioral mechanisms, and class-level effects.

1. Metric Conclusions

- Utilization is not an access metric under oversubscription; it measures completed visits per slot.
- Served rate is the primary aggregate access metric; it measures completed visits per arrival.
- Offered wait must be reported with no-offer share because no-offer patients are absent from wait metrics.

2. Capacity Pressure

- Increasing arrivals lowers served rate and raises offered wait when capacity is fixed.
- At the stress-test baseline ($\lambda = 100$, $S = 32$), high utilization (0.841) coexists with poor access (0.269).
- The stress-test baseline is useful for mechanism diagnosis, not as a calibrated clinic estimate.

3. Behavioral Mechanisms

Mechanism	Main effect	Interpretation
Cancellation	Reduces own-class served rate.	May free future slots for other patients.
Balking	Reduces own-class served rate.	Reallocates access through the shared FCFS pool.
No-show	Directly lowers utilization.	Missed slots are not rebooked.

4. Class-Level Conclusions

- With symmetric parameters, the expected class access gap is near zero under pooled FCFS.
- Class gaps emerge from asymmetric behavioral parameters or arrival rates, not explicit class priority.
- In the realistic scenario, aggregate outcomes improve but the served-rate gap remains substantial (0.220), so class-level metrics must be reported.

5. Reporting Recommendation

Do not summarize the model with utilization alone.

Minimum reporting set:

- served rate;
- utilization;
- offered wait;
- no-offer share;
- outcome decomposition;
- class-level served rates.

6. Meeting Decision

Meeting decision: choose the reference case for future analysis: stress-test baseline for mechanism diagnosis, or calibrated realistic scenario for operational interpretation.